Chap 2

Date No.

the do statement

the control expression is evaluated after each execution of the controlled statement

eg.
```
n=1;
do
{
  printf("%d", n);
  n=n+1;
}
while(n<=10);
```

switch & break

eg.
```
scanf("%d", &command);
switch(command)
{
case 1: .... break;
  :
default: ... break;
}
```

If there's no break, execution will continue from the matched case label to the end of the switch statement

the case label $\boxed{\text{default}}$: corresponds to any value of the control expression that is not represented by another case label

eg.
```
int price(char label)
{
  switch(label)
  {
    case 'A': case 'B': case 'C': return 9;
    case 'D': case 'E': return 15;
  }
}
```

break in a switch : exit only the switch

continue : skip the rest of the loop body & start next loop iteration

the conditional operator

exp-c $\boxed{?}$ exp-t $\boxed{:}$ exp-f ;

control true false

exponential function

```
# include <math.h>
exp(x)
```

the math library functions are defined to take a double as their input & output

break

the break statement can be used ~~to~~ in terminating all 3 loops for, while & do ⋯ while loops. The execution will come out from the immediate layer of iteration.

It only breaks out of one loop, not out of a conditional statement

① while ( )
```
while (   )
{
   ....
   if (   )
   {
      ...
      break;
   }
}
```

② do
```
do
{
   ...
   if(   )
   {  ...
      break;
   }
}
while (..);
```

③ for(..,..,..)
```
for(..,..,..)
{
   ...
   if(   )
   {
      ...
      break;
   }
}
```

continue;

① while (
```
while (   )
{
   ...
   if(   )
   {
      ...
      continue;
   }
}
```

② do
```
do
{
   ...
   if(   )
   {
      ...
      continue;
   }
}
while(..);
```

③ for(..;..;..)
```
for(..;..;..)
{
   ...
   if(   )
   {
      ...
      continue;
   }
}
```

random number

\# include <stdlib.h>

srand(seed); → use prime num. to decrease the possibilities of repetition

rand() → [0, 32767] returns integer

\# include <time.h>

srand(time(NULL));

⊗ if the seed is e.g., once at the beginning. the sequence of numbers produced by rand() will always be the same every time you run the program.

Same seed follows the same deterministic algorithm.

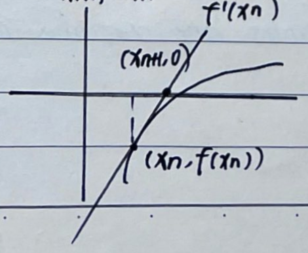
macro

1. Always put ( ) around each parameter inside the macro body
2. Usually, also put ( ) around the whole expression.

root

establishing the equation by setting the slope equal.

$$f'(x_n) = \frac{0 - f(x_n)}{x_{n+1} - x_n} \rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



$$G(x_n) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} \rightarrow x_{n+1} = x_n - \frac{f(x_n)}{G(x_n)}$$